

SADHWIKA RACHAMALLA

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SUMMARY

Software Development Engineer with 2+ years of experience building cloud-native, full-stack applications using Java, Python, React.js, and AWS. Proficient in designing scalable, distributed systems and high-performance RESTful APIs. Experienced in CI/CD, object-oriented design, and modern software engineering practices including serverless and containerized architectures.

EDUCATION

Master of Engineering in Computer Science, University of Cincinnati, Ohio

August 2023- May 2025

SKILLS

Languages: Java, Python, C++, C#, C, HTML, CSS

Frameworks and Libraries: Spring Boot, Express.js, React.js, Node.js, FastAPI, Tailwind CSS

Cloud and DevOps: AWS, Azure, Docker, Kubernetes, Jenkins, Terraform, GitHub Actions

Databases, Data Technologies and Operating Systems: SQL, MySQL, MongoDB, Redis, Kafka, Windows, Linux

Practices and Tools: Git, Jira, Agile/Scrum, CI/CD, Infrastructure as Code, TensorFlow, PyTorch, Grafana, ELK Stack, PyTest, Figma

EXPERIENCE

Software Engineer, Bitbyte Solutions, Remote

December 2024- Present

- Designed and programmed a document classification and processing system for automating metadata extraction and reporting using **Python**, **FastAPI**, and **Kafka**, improving system throughput by 15%.
- Implemented **RESTful APIs** for scalable metadata services, reducing latency by 20% and increasing data throughput to support 12+ microservices and analytics tools.
- Utilized **DynamoDB** with indexing strategies to enhance query speed, reducing backend load during traffic spikes and boosting data access for 12+ consumer endpoints.
- Containerized services using **Docker** and configured **GitHub Actions** for CI/CD, increasing deployment reliability.
- Set up **Grafana** dashboards and integrated **CloudWatch** logs to monitor system performance, reducing mean time to resolution and improving customer retention by 25%.

Software Engineer-Graduate Student Assistant, University of Cincinnati, Ohio

September 2023- December 2024

- Built backend components using **Spring Boot** and **MySQL**, enhancing secure access and data integrity for 100+ system users.
- Engineered scalable, role-based **REST APIs** that improved system responsiveness and reduced data retrieval time by 20%.
- Styled frontend using **React**, **Tailwind CSS** and **Chakra UI** to improve accessibility and mobile responsiveness, reducing UI issues by 30% and improving user feedback.
- Presented new features to academic teams, gathering actionable feedback and driving system-wide adoption.

Software Developer, Forge Alumnus, India

March 2021- April 2023

- Engineered a real-time inventory tracker using **Java** for logic, **NestJS** for backend APIs, and **Next.js** for dashboard UI, reducing inventory mismatches by 30% and enhancing forecast reliability.
- Integrated **MongoDB** for flexible data storage and **Redis** for low-latency caching, enhancing real-time data access and improving dashboard responsiveness for 150+ active users.
- Designed low-latency **REST APIs** with average response time under 200ms, enabling real-time inventory sync across systems.
- Investigated and resolved production bugs using **performance profiling** and **logging tools**, reducing downtime by 20%.
- Participated in **Agile** contributions and facilitated smoother team execution using **JIRA** to drive consistent release cycles.

PROJECTS

Potato Crop Disease Prediction using Deep Learning | [GitHub](#) | [Paper](#)

- Initiated a **CNN**-based image classification model using **TensorFlow** and **data augmentation** to detect potato plant diseases with 96.8% accuracy, aiding early diagnosis for farmers.
- Deployed the optimized model via **FastAPI** and **TensorFlow Serving**, integrated with a **React/React Native** frontend and **Google Cloud Functions** for real-time mobile predictions in low-resource regions.

Cell City Adventure – Educational Web Game | [GitHub](#)

- Created an interactive game for school students featuring clickable cell maps, animated quizzes, and voice-based fun facts.
- Engineered a responsive UI using **React**, **Node.js/Express.js** for backend logic, integrating **MongoDB** for storing quiz progress and **Bootstrap** for responsive design, enhancing student engagement.

CERTIFICATIONS

AWS Certified Solutions Architect-Associate | [Link](#)

Microsoft Certified: Azure Fundamentals | [Link](#)